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Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Introduction to Networks

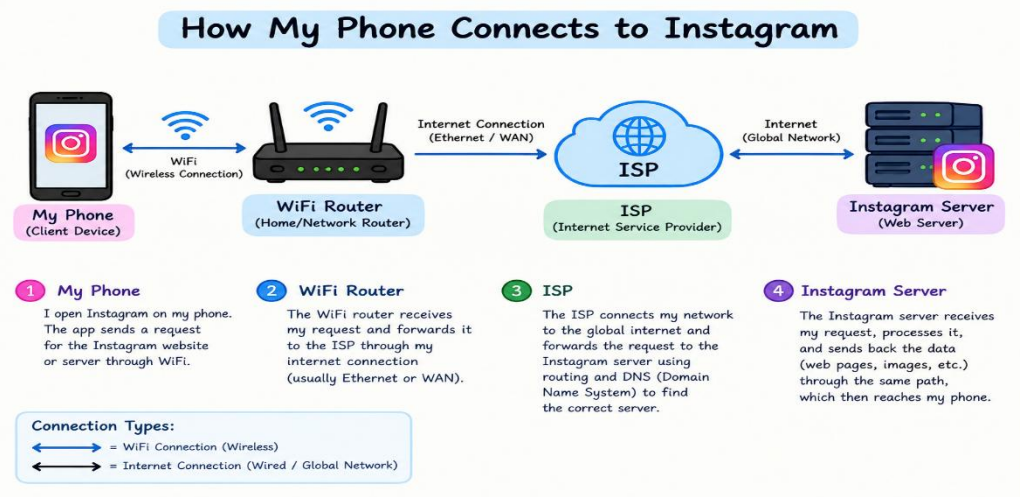
Task:

❖ **1. Draw a simple diagram (hand-drawn or digital) showing how your phone connects to Instagram, including your device, WiFi router, ISP, and Instagram server. Label each part and the connections.**

➢ **Aim**

- To understand how a mobile phone connects to Instagram through a Wi-Fi router and Internet Service Provider (ISP), and to identify the main components involved in network communication.

➢ **Diagram: How a Phone Connects to Instagram**



➢ Components and Their Functions

Component	Function
Mobile Phone	Runs the Instagram app and sends requests to access content.
Wi-Fi Router	Connects the phone to the local network and forwards traffic towards the Internet.
ISP (Internet Service Provider)	Provides Internet access and routes data between the user's network and the Internet.
Internet	Connects networks and carries data between the user's device and remote servers.
Instagram Server	Processes requests and sends back content such as posts, reels, images and messages.

➢ Working Procedure

1. Mobile Phone: I open the Instagram application on my phone.
2. Wi-Fi Connection: The phone sends data wirelessly to the Wi-Fi router.
3. Wi-Fi Router: The router forwards the data towards the Internet through the ISP.
4. ISP and Internet: The ISP provides connectivity, and network routing carries the request towards Instagram's servers. DNS helps resolve the domain name to an IP address when needed.
5. Instagram Server: The server processes the request and sends the required content back.

6. Receiving Data: The response travels back through the Internet, ISP and Wi-Fi router to my phone, where Instagram displays the content.

➢ **Types of Connections Used**

- Wireless connection: Between the mobile phone and Wi-Fi router using Wi-Fi.
- Wired or other access connection: Between the router and ISP, depending on the Internet service.
- Internet connection: Between the ISP's network and Instagram's servers through the Internet.

➢ **Real-World Example**

➢ When I open Instagram and watch a reel, my phone sends a request through the Wi-Fi router and ISP. The request travels across the Internet to Instagram's servers. The server sends the required video data back to my phone, and the reel appears on the screen.

➢ **Result**

➢ The network diagram was prepared to demonstrate how a mobile phone connects to Instagram through a Wi-Fi router, Internet Service Provider and the Internet. The functions of the main components and the path followed by data were studied.

❖ **2. List 5 devices you use or see daily that are part of a network (for example: smartphone, smart TV, WiFi router, laptop, smartwatch). For each, write what type of network it connects to (LAN, WAN, PAN, etc.).**

➢ **Aim**

- To identify five devices used in daily life that are part of a network and understand the types of networks they connect to.

➢ **Introduction**

- A **computer network** is a group of interconnected devices that communicate and share information or resources. Different devices connect to different types of networks, such as a Personal Area Network (PAN), Local Area Network (LAN), and Wide Area Network (WAN).

➢ **Five Network Devices Used in Daily Life**

Sr. No.	Device	Type of Network	Explanation
1	Smartphone	LAN / WAN / PAN	A smartphone connects to a home Wi-Fi network (LAN), accesses the Internet through mobile data (WAN), and connects to wireless accessories using Bluetooth (PAN).
2	Laptop	LAN / WAN	A laptop connects to a local network through Wi-Fi or Ethernet and accesses websites and online services through the Internet.

3	Smart TV	LAN / WAN	A smart TV connects to the home Wi-Fi network (LAN) and accesses online services such as YouTube and Netflix through the Internet (WAN).
4	Wi-Fi Router	LAN / WAN	A Wi-Fi router connects devices within a local network (LAN) and forwards their Internet traffic through the ISP's network (WAN).
5	Smartwatch	PAN	A smartwatch often connects to a smartphone through Bluetooth, forming a Personal Area Network (PAN). Some smartwatches can also connect directly through Wi-Fi or cellular networks.

- **Types of Networks**

  1. PAN (Personal Area Network): A small network connecting personal devices over a short distance, such as a smartphone and smartwatch connected through Bluetooth.

  2. LAN (Local Area Network): A network connecting devices within a limited area, such as a home, office, school or computer laboratory.

  3. WAN (Wide Area Network): A network that connects devices and smaller networks over large geographical areas. The Internet is the most common example of a WAN.

- **Real-World Example**
- At home, a laptop and smart TV connect to the Wi-Fi router through a LAN. The router forwards their Internet traffic through the ISP's network. A

smartphone may also connect to a smartwatch through Bluetooth, which is an example of a PAN.

- **Result**
- Five commonly used network devices were identified, and their network connections were explained using PAN, LAN and WAN.

❖ **3. Explain the difference between the Internet and an intranet using an example from your college or a public WiFi you have used.**

- **Aim**
  - To understand the difference between the Internet and an intranet with the help of a real-world example from a college network.

- **Introduction**
  - The **Internet** and an **intranet** are both networks used for communication and sharing information. However, they differ in their accessibility, purpose and users.

- **Internet**
  - The **Internet** is a worldwide network of interconnected computer networks. It allows people to access websites, communicate online and share information across the world.
  - **Examples:** Google, YouTube, Instagram and Wikipedia.

- **Intranet**

- An **intranet** is a private network used within an organization, such as a college, school or company. It allows authorized users to access internal information, applications and resources.
- **Examples:** A college's internal student portal, staff information system or private internal file-sharing system.

➢ **Difference Between Internet and Intranet**

Internet	Intranet
It is a global network available worldwide.	It is a private network used within an organization.
It can be accessed by anyone with a suitable Internet connection, subject to service restrictions.	It is accessible only to authorized users.
It provides public websites and online services.	It provides internal information and services.
It connects networks across the world.	It connects users and resources within an organization.
Example: Visiting Google or Instagram.	Example: Accessing a college's private student portal.

- **Real-World Example from a College**
- Suppose I am using the Wi-Fi network in my college.
  - **Internet:** When I connect my phone to the college Wi-Fi and open Google, YouTube or Instagram, I am accessing services through the Internet.
  - **Intranet:** If my college has a private student portal available only to authorized students and staff, that portal is an example of an intranet service. It may provide attendance records, internal notices, timetables or study materials.
- A college Wi-Fi network does not automatically become an intranet just because it is private or password-protected. An intranet refers to the organization's private network and internal resources.

❖ **4. Pick any one app you use (like WhatsApp, Zomato, or Flipkart) and describe, in 4-5 lines, how data travels from your device to their server and back when you perform a search inside the app. Mention the devices and networks involved, and what happens when you hit 'search'.**

➢ **Application Selected: Flipkart**

1. When I open the Flipkart app on my smartphone and enter a product name in the search bar, the app prepares a search request.
2. The request travels from my smartphone through the Wi-Fi router or mobile network to my Internet Service Provider (ISP).
3. Through the Internet, the request reaches Flipkart's servers, where it is processed to find relevant products.
4. The server sends the search results back through the Internet, ISP and Wi-Fi router or mobile network to my smartphone.
5. Finally, the Flipkart app displays the matching products on my screen. This process involves network devices, IP addressing, and secure communication protocols such as HTTPS.